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LECTURER: DR. ANN KIARIA

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GROUP 2 REPORT

Prepared by:

YUSSUF ELMI – GROUP LEADER – BCS-1-3008-2/2025
AGENG'O ELIZABETH ANYIM MULA – COM-1-6271-3/2024
NHIAL NYALAAM DUOTH – BIR-1-2504-2/2025
SAHRA ABDALA ADAN – BCS-1-2519-2/2025
ABDULLAHI OMAR HASSAN

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Declaration

This research report was prepared collectively as a group assignment. We hereby declare that this report is the original work of **Group 2** and has not been submitted elsewhere for academic credit. All contributors participated in the fieldwork, data collection, analysis, and preparation of the final report.

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Abstract

This report examines the urban built environment and land use planning in Nairobi, with particular focus on land use patterns, zoning practices, building density, and the availability of green spaces within selected urban areas. The study aimed to evaluate the extent to which Nairobi's urban form reflects sustainable urban planning principles and to identify challenges associated with urban sprawl.

Data for the study were collected through field observation, visual assessment, and the use of aerial photographs from online sources. The findings indicate that Nairobi is characterized by high building density, widespread mixed land use, and limited green spaces in many parts of the city. The study also observed evidence of weak zoning enforcement, especially in areas where commercial and residential functions coexist without clear regulation.

The findings are relevant to the *Environmental Management and Coordination Act (EMCA)*, the *Climate Change Act*, and *Kenya Vision 2030*, all of which advocate for sustainable development. The report also supports *Sustainable Development Goal 11*, which seeks to make cities inclusive, safe, resilient, and sustainable.

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1. Introduction

Rapid urbanization has significantly influenced the physical and environmental development of Nairobi. This growth has resulted in the expansion of built-up areas and increasing pressure on environmental resources. As the city's population continues to grow, the demand for housing, infrastructure, transport systems, and healthy living conditions also increases. To address these demands effectively, Nairobi must promote self-sustainability through proper urban planning.

Urban planning plays a vital role in ensuring that cities develop in a sustainable and organized manner. Proper land use planning promotes efficient utilization of available space, reduces environmental degradation, improves access to services, and enhances the overall quality of life for residents. It also promotes inclusivity by ensuring that commercial, residential, institutional, and infrastructural developments are well integrated.

This report presents findings from a field study conducted in Nairobi. The study involved direct observation, visual assessment, and analysis of aerial images obtained from online sources.

Objectives of the Study

The objectives of the study were:

- To observe building density and land use patterns in Nairobi.
- To identify evidence of zoning or lack of zoning.
- To assess the availability of green spaces within built-up areas.
- To examine how Nairobi reflects sustainable urban planning principles.
- To identify the challenges of urban sprawl.
- To relate the findings to environmental policies and Sustainable Development Goal 11.

2. Methodology

This study was conducted through field observation in selected areas of Nairobi. The areas studied included Nairobi CBD, Westlands, Kayole, Karen, Langata, and Ruaka. These areas were selected because they represent commercial zones, residential neighborhoods, low-density suburbs, and mixed-use developments.

Data were collected through direct observation of the physical environment and the use of aerial images from various online sources. Observations focused on building density, land use patterns, zoning characteristics, and the availability of green spaces within built-up environments.

Building density was assessed by observing the spacing between buildings and the number of floors in typical structures. Areas with closely spaced multi-storey apartment buildings were classified as high-density areas, while areas with fewer and more widely spaced buildings were classified as low-density areas.

Land use patterns were identified by observing the types of activities taking place in different locations. These included residential activities, commercial businesses, office developments, and public facilities such as schools and religious institutions.

Evidence of zoning was identified by examining whether similar types of land use were grouped together or whether different activities occurred within the same area. Green spaces were assessed by observing the presence of parks, trees, open spaces, and undeveloped land within built-up sections of the city.

The data collected were analyzed qualitatively in order to identify patterns and environmental conditions within the selected study areas.

3. Findings

3.1 Building Density

The study revealed varying levels of building density across different parts of Nairobi. Commercial areas showed the highest building density, with multi-storey buildings located close to one another and minimal spacing between structures. Such areas included Nairobi CBD and commercial sections of Westlands.

Residential areas showed moderate building density, especially in planned estates where houses and apartment buildings were arranged in relatively organized patterns. Examples of moderate-density residential areas included South B, South C, and Langata.

In some areas, however, buildings were very closely spaced due to the increasing demand for housing. Such high-density residential areas included Pipeline, residential parts of Eastleigh, and Kayole.

Other areas showed low building density, particularly high-income suburbs such as Karen, Runda, and Gigiri, where buildings were fewer and more widely spaced.

While high density can contribute to efficient land use, it may also increase environmental pressure through congestion and increased demand for services such as water supply, waste management, and transport infrastructure. In addition, overcrowding can reduce air circulation and ventilation within urban neighborhoods.

3.2 Land Use Patterns

Mixed land use patterns were widely observed in the study areas. Commercial activities such as shops, kiosks, and small businesses were commonly located within residential neighborhoods. In some buildings, commercial activities were situated on the ground floor while residential units occupied the upper floors.

The presence of mixed land use allows residents to access services more easily without travelling long distances. This reduces transport costs, saves time, and supports local economic activity within neighborhoods.

However, unplanned mixed land use may also result in congestion, noise pollution, and other

environmental problems, especially where commercial activities expand into residential areas without proper planning and regulation.

3.3 Zoning Practices

Limited evidence of strict zoning regulations was observed in several areas. In some residential neighborhoods, commercial businesses such as retail shops, workshops, and service outlets were operating alongside housing developments. This indicates weak enforcement of zoning regulations and reflects unplanned urban growth.

Proper zoning is important because it helps separate incompatible land uses, reduces environmental conflicts, and improves the overall living conditions of residents. Some planned estates showed evidence of better zoning through consistent building types, organized layouts, and more controlled development patterns.

Despite the existence of a formal zoning framework under the *Physical and Land Use Planning Act (2019)*, the findings suggest that although zoning laws exist, enforcement mechanisms remain weak in many parts of Nairobi.

3.4 Green Spaces

The study showed limited availability of green spaces in many built-up areas. Only a few public parks, forest reserves, and recreational open spaces were clearly observed. These included Uhuru Park, Karura Forest, Nairobi Arboretum, and Ngong Road Forest Sanctuary.

It was also evident that green spaces are unevenly distributed across the city. Some neighborhoods had very few trees, parks, or open recreational areas, particularly in highly built-up and densely populated zones.

Green spaces play a critical role in major cities such as Nairobi. They improve air quality, reduce urban heat, provide recreational areas, support biodiversity, and contribute to the mental and physical well-being of residents. Therefore, although large green spaces such as Uhuru Park and Karura Forest offer significant environmental and recreational benefits, their limited and uneven distribution remains a challenge.

4. Discussion

4.1 How Nairobi's Urban Form Reflects Sustainable Urban Planning Principles

Nairobi's urban form reflects some principles of sustainable urban planning, particularly through compact development and mixed land use patterns. In many parts of the city, residential areas are located close to commercial centers, allowing residents easier access to services such as shops, schools, and workplaces. This reduces travel distances and transport costs.

High building density in areas such as Nairobi CBD and Eastleigh also demonstrates efficient land use. Compact development can help reduce urban sprawl and promote better use of limited land resources.

However, sustainable urban planning in Nairobi remains limited by the shortage and uneven distribution of green spaces, as well as weak zoning enforcement in some areas. Sustainable urban development requires balanced growth that integrates environmental protection, proper zoning, infrastructure development, and access to public open spaces. These elements are not fully achieved in many parts of Nairobi.

4.2 Challenges of Urban Sprawl in Nairobi

Urban sprawl is a major challenge in Nairobi due to rapid population growth and increasing demand for housing. The outward expansion of the city into surrounding areas has increased the distance between residential neighborhoods and workplaces, resulting in longer commuting times and higher transport costs.

Urban sprawl has also placed growing pressure on infrastructure such as roads, water supply systems, drainage systems, and electricity networks. In addition, the expansion of built-up areas often leads to the loss of agricultural land, wetlands, and natural vegetation. This contributes to environmental degradation and makes urban development less sustainable.

The continued spread of unplanned development also increases climate vulnerability, particularly where natural drainage systems are disrupted, thereby increasing the risk of flooding and other environmental hazards.

4.3 Relevant Provisions of the Environmental Management and Coordination Act (EMCA)

Several provisions of the *Environmental Management and Coordination Act (EMCA)* are relevant to urban development. First, EMCA requires that Environmental Impact Assessments (EIAs) be conducted before major development projects such as roads, buildings, and industrial activities are undertaken. This ensures that environmental consequences are considered before implementation begins.

The Act also promotes environmental conservation, pollution control, and sustainable land use planning. In addition, it requires environmental audits to monitor ongoing development projects and ensure compliance with environmental standards.

These provisions are important because they help reduce the negative environmental impacts of urban development. However, the findings of this study suggest that implementation remains inconsistent. In many cases, environmental laws appear to exist in formal policy documents, but practical enforcement remains weak.

4.4 Link to Sustainable Development Goal 11

The findings of this study relate closely to *Sustainable Development Goal 11*, which aims to make cities inclusive, safe, resilient, and sustainable. Mixed land use development and compact building patterns support efficient use of land and can contribute positively to the development of sustainable cities.

However, limited green spaces, weak zoning enforcement, and unplanned development indicate significant challenges in achieving sustainable urban environments in Nairobi. Improved planning, stronger policy implementation, better environmental protection, and increased investment in green public spaces would help Nairobi move closer to achieving SDG 11.

The study also has implications for climate action because urban planning directly affects environmental sustainability, resilience, and the city's ability to respond to climate-related risks.

5. Conclusion

The study concludes that Nairobi exhibits a complex urban form characterized by varying building densities, widespread mixed land use, weak zoning enforcement, and limited green spaces in many built-up areas. While some aspects of Nairobi's development reflect sustainable urban planning principles, particularly compact development and accessibility of services, major gaps remain in environmental protection and planning enforcement.

Urban sprawl, inadequate green space distribution, and inconsistent implementation of planning laws continue to undermine the sustainability of the city. To improve urban development outcomes, stronger zoning enforcement, increased protection of green spaces, and more integrated planning approaches are necessary.

Overall, sustainable urban planning is essential for ensuring that Nairobi develops in a way that is inclusive, resilient, environmentally sound, and aligned with both national development goals and global sustainability frameworks.

Reference List

1. Government of Kenya. (1999). *Environmental Management and Coordination Act*.
2. Government of Kenya. (2016). *Climate Change Act*.
3. Government of Kenya. (2007). *Kenya Vision 2030*.
4. Nairobi City County. (2014). *Nairobi Integrated Urban Development Master Plan*.
5. UN-Habitat. (2020). *The State of African Cities*.
6. United Nations. (2015). *Transforming Our World: The 2030 Agenda for Sustainable Development*.